

High-Molecular-Weight gDNA extraction protocol from moss gametophytes

Lei Zhu

Department of Botany and Plant Sciences, University of California, Riverside

Email: lzhu002@ucr.edu

Abstract

The purpose of this protocol is to extract high-molecular-weight genomic DNA from gametophyte tissue of the moss *Physcomitrella patens*, which contains most of the identified phenolic compounds. This protocol was modified from a CTAB method for DNA isolation from moss protonemal tissue (Ano et al., 2012). gDNA extracted using this modified CTAB protocol is pure, comparable to gDNA quality isolated from commercial extraction kits. Besides, the yield (69 µg/g FW) is relatively high. The high-quality gDNA is useful for PCR and Southern blotting for the detection of genome editing.

Keywords: high-molecular-weight, genomic DNA extraction, *Physcomitrella patens*

Before Start

1. Prepare reagents as listed
2. Fill liquid nitrogen in a small thermal tank
3. Set water bath to 65°C
4. Set incubator to 37°C

Reagents

- 2x CTAB buffer: 2% CTAB (Hexadecyltrimethylammonium Bromide), 1.4M NaCl, 100mM Tris-HCl pH 8.0, 20mM EDTA

OPTIONAL: 1% PVP-40 can be added to 2x CTAB buffer for removing phenolic compound in plant extract.

- 10% CTAB: 10% CTAB, 0.7 M NaCl
- Chloroform:isoamyl alcohol (24:1)
- Tris-saturated Phenol (PH > 7.5),
- Phenol:chloroform (25:24),
- 100% Ethanol,
- 70% Ethanol,
- 10 mg/mL RNase A,
- 3M Sodium Acetate (NaOAc, pH 5.2)
- TE: 10mM Tris-HCl, 1mM EDTA, PH 8.0

Materials

- 3-week-old moss gametophytes, 2-5 grams of fresh weight (FW)

Equipment

- Mortar and pestle
- Liquid nitrogen
- 50 mL conical tube (sterile, DNase free)
- Microcentrifuge tubes, 1.5 mL (sterile, DNase free)
- Desktop centrifuge for 50 mL conical tubes
- Microcentrifuge
- Water bath
- Incubator

Procedures

A. Harvest Plant Materials

1. Harvest young gametophyte tissue that was cultured from protonema tissue for three weeks. Carefully remove rhizoids from leafy gametophytes. Dry leafy gametophytes with clean towel papers.

IMPORTANT: Rhizoids mixed in the green leafy tissue will cause pigment co-precipitated with DNA. Be careful to remove rhizoids from leafy gametophytes.

2. Weigh 2–5 g fresh weight of leafy gametophytes and grind plant tissue to a fine powder with mortar and pestle in liquid nitrogen.

IMPORTANT: Grind moss leafy gametophytes to a fine powder is very important. If the powder is crude, the efficiency of extraction will be much lower than expected.

3. Transfer the ground powder to a 50 ml conical tube. Open the cap and put the tube in a -80°C freezer until all liquid nitrogen evaporated.

B. Crude Extraction in CTAB Buffer

4. Add 10 mL pre-heated 2x CTAB buffer to a 50 mL conical tube and cap it. Shake the tube well to mix CTAB buffer and ground powder. Incubate tube in 65°C water bath for 45 min. Shake the tube to mix buffer and plant powder every 10 min.
5. Take the conical tube out of the water bath and cool down to room temperature. Add 10 mL chloroform: isoamyl alcohol to the conical tube. Then put the tube to a rotator for 20 min at room temperature.
6. Centrifuge at 1,000 g for 5 min in room temperature to separate phase.
7. Transfer supernatant to a new tube, add 1mL 10% CTAB, and 10 ml chloroform: isoamyl alcohol. Incubate the mixture at room temperature for 5 min and centrifuge at 1,000 g for 5 min to separate the phase.
8. Transfer aqueous supernatant to a new conical tube, adding 10 mL isopropanol, and incubate on ice for 10 min.
9. Centrifuge at 1,000 g for 5 min to precipitate pellet.
10. Decant supernatant, air dry the pellet.
11. Dissolve the pellet in 4 mL TE buffer. (Tap the pellet gently until it dissolves completely)
12. Centrifuge at 1,000 g for 5 min to remove any impurities.
13. Transfer supernatant to a new microtube, add 400 µL 3M sodium acetate, 8 ml 100% pre-chilled ethanol. Incubate on ice for 20 min.

14. Centrifuge at 1,000 g for 5 min to precipitate pellet.
15. Rinse pellet with 70% ethanol.
16. Air dry pellet. Dissolve pellet in 700 μ L TE. (Tap the pellet gently. Wait until the pellet is completely dissolved. DNA can be stored at this step for overnight at 4°C)
17. Add RNase A into TE-dissolved DNA solution at a final concentration of 50 μ g/mL. Incubate at 37°C for 30 min.

C. Removal of protein

18. Add 700 μ L Tris-saturated Phenol. Mix thoroughly.
19. Centrifuge at 16,000 rcf for 5 min at room temperature to separate phases. Transfer upper phase to a new tube. Leave white stuff in the middle layer between two phases.
20. Extract in phenol twice. Repeat steps 18-19.
21. Transfer the upper aqueous phase to a new microtube. Add an equal volume of phenol: chloroform. Mix thoroughly.
22. Centrifuge at 16,000 rcf for 3 min in room temperature to separate phases. Transfer upper phase to a new tube.
23. Transfer the upper phase to a new microtube. Add 700 μ L chloroform: isoamyl alcohol. Mix thoroughly.
24. Centrifuge at 16,000 rcf for 3 min at room temperature to separate phase. Transfer aqueous phase to a new tube.
25. Repeat steps 23-24 once to remove phenol residue.

D. Precipitation of Genomic DNA

26. Transfer supernatant to a new microtube. Add 1/10 volume of 3 M sodium acetate and 2x volume of pre-chilled 100% ethanol to the supernatant. Incubate on ice for 10 min.
27. Centrifuge at 16,000 rcf for 5 min to collect pellet.
28. Wash pellet with 70% ethanol. Centrifuge down the pellet at 16,000 rcf for 5 min.
Pellet may not be attached tightly to the bottom of the microtube. Be careful to pour off supernatant.
29. Air-dry pellet.
30. Dissolve pellet in 100 μ L TE or ddH₂O.

Additional Notes

1. The entire procedure takes about 4 hours.
2. The yield and quality of gDNA from four preps are listed below. FW, fresh weight.

Plant Tissue	Extraction Method	Yield (μ g/g FW)	OD 260/280	OD 260/230
WT- Gametophytes	The modified CTAB method	69.21 $\pm$ 8.43	1.95 $\pm$ 0.03	2.21 $\pm$ 0.12
WT- Gametophytes	NucleoSpin DNA extraction Kit	Plant 5.81 $\pm$ 0.70	1.99 $\pm$ 0.02	2.20 $\pm$ 0.05

Citation

Ano, N., Sato, Y., Nishiyama, T. 2012. Genomic DNA extraction. In: PHYSCOmanual ver2.0. <http://moss.nibb.ac.jp/protocol.html>